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References:

1. Engineering Mathematics – D.C. Gokhroo & others, Unique Publications
2. Advanced Engineering Mathematics – Erwin Kreyszig, John Wiley & Sons
3. Advanced Engineering Mathematics – R.K. Jain & S.R.K. Iyengar, Narosa Publications
4. Higher Engineering Mathematics – B.V. Ramana, McGraw Hill Education
5. Advanced Engineering Mathematics – H.K. Das, S. Chand Pub.
6. Engineering Mathematics – Srimanta Pal & Subodh C. Bhunia, Oxford University Press
7. Engineering Mathematics – A.R. Vashistha & Vipin Vashistha, Pragati Prakashan

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UNIT-III (INTEGRAL CALCULUS)

LECTURE NO.- 19

Beta and Gamma Functions

Beta and Gamma functions are very useful special functions (or higher functions), which are expressed as definite integrals. Many integrals which can not be expressed in terms of elementary functions (algebraic and transcendental functions) can be evaluated in terms of beta and gamma functions.

Definition of Beta Function: Euler's First Integral

For positive values of m and n , an improper definite integral $\int_0^1 x^{m-1}(1-x)^{n-1}dx$ is called beta function and expressed by the notation $\beta(m, n)$, where m and n are the parameters of the integral. It is also called Euler's integral of the first kind.

$$\therefore \beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx, m > 0, n > 0$$

Definition of Gamma Function: Euler's Second Integral

For positive value of n , an improper definite integral $\int_0^\infty e^{-x}x^{n-1}dx$ is called gamma function and expressed by the notation Γn , where n is the parameter of the integral. It is also called Euler's integral of the second kind.

$$\therefore \Gamma n = \int_0^\infty e^{-x}x^{n-1}dx, n > 0$$

Properties of Beta Function

1. Symmetry of beta function: $\beta(m, n) = \beta(n, m)$.

Proof. By the definition of the beta function, we have

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx, m > 0, n > 0 \dots\dots\dots (1)$$

Putting $x = 1 - y$ in (1), we get

$$\beta(m, n) = \int_1^0 (1-y)^{m-1}(y)^{n-1}(-dy), \therefore \text{limit}$$

x	0	1
$y = 1 - x$	1	0

$$= \int_0^1 y^{n-1} (1-y)^{m-1} dy, \text{ by a property of definite integral}$$

$$= \int_0^1 x^{n-1} (1-x)^{m-1} dx, \text{ by a property of definite integral}$$

$$= \beta(n, m), \text{ comparing with (1).}$$

$$\therefore \boxed{\beta(m, n) = \beta(n, m)}$$

2. Trigonometric representation of beta function:

$$\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0.$$

Proof. By the definition of the beta function, we have

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, m > 0, n > 0 \dots\dots\dots (1)$$

Putting $x = \sin^2 \theta$ in (1), we get

$$\begin{aligned} \beta(m, n) &= \int_0^{\frac{\pi}{2}} \sin^{2m-2} \theta \cdot \cos^{2n-2} \theta \times 2 \sin \theta \cdot \cos \theta d\theta, \because \text{limit} \\ &= 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta \end{aligned}$$

x	0	1
θ	0	$\pi/2$

$$\therefore \boxed{\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0.}$$

3. Beta function in the form of an improper integral of first kind:

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx, m > 0, n > 0.$$

Proof. By the definition of the beta function, we have

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, m > 0, n > 0 \dots\dots\dots (1)$$

$$\text{Let } x = \frac{1}{1+y}. \text{ So } dx = -\frac{1}{(1+y)^2} dy \text{ and } 1+x = \frac{1}{y} \Rightarrow y = \frac{1}{x} - 1 \text{ for limit of } y.$$

$$\text{Also } 1-x = 1 - \frac{1}{1+y} = \frac{y}{1+y}.$$

Putting these values in (1), we get

$$\beta(m, n) = \int_{\infty}^0 \frac{1}{(1+y)^{m-1}} \frac{y^{n-1}}{(1+y)^{n-1}} \left\{ -\frac{1}{(1+y)^2} \right\} dy, \because \text{limit}$$

x	0	1
$y = \frac{1}{x} - 1$	∞	0

$$= \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy, \text{ by a property of definite integral}$$

Again using a property of definite integral, we have

$$\beta(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx \dots\dots\dots (2)$$

Interchanging m and n in (2), we get

$$\beta(n, m) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{n+m}} dx \dots\dots\dots (3)$$

By symmetry, we have

$$\beta(m, n) = \beta(n, m)$$

$$= \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{n+m}} dx \text{ by (3)}$$

$$\therefore \boxed{\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx, m > 0, n > 0.} \dots\dots\dots (4)$$

Note 1. From (4) and (2) of above proof, we have obtained

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0.$$

Note 2. Adding (4) and (2) of above proof, we have

$$\begin{aligned} 2\beta(m, n) &= \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx \\ &= \int_0^{\infty} \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx \\ \Rightarrow \beta(m, n) &= \frac{1}{2} \int_0^{\infty} \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0. \end{aligned}$$

$$4. \beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0.$$

Proof. From property 3, we have

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx, m > 0, n > 0$$

Using a property of definite integral, we have

$$\beta(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_1^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx \dots\dots\dots (1)$$

Putting $x = \frac{1}{y}$ in the second integral of R.H.S. (1), we get

$$\begin{aligned} \int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx &= \int_1^0 \frac{1}{y^{m-1} \left(1+\frac{1}{y}\right)^{m+n}} \left(-\frac{1}{y^2}\right) dy, \because \text{limit} \\ &= \int_0^1 \frac{1}{y^{m-1} \left(\frac{y+1}{y}\right)^{m+n}} \frac{1}{y^2} dy, \text{ by a property of definite integral} \\ &= \int_0^1 \frac{y^{m+n}}{y^{m-1}(1+y)^{m+n}y^2} dy \\ &= \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}} dy \\ &= \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx, \text{ by a property of definite integral} \end{aligned}$$

Putting it in (1), we get

$$\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0.$$

$$5. \beta(m, n) = a^m b^n \int_0^\infty \frac{x^{m-1}}{(ax+b)^{m+n}} dx, m > 0, n > 0.$$

Proof. From property 3, we have

$$\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx, m > 0, n > 0 \dots\dots\dots (1)$$

Putting $x = \frac{ay}{b}$ in (1), we get

$$\begin{aligned} \beta(m, n) &= \int_0^\infty \frac{a^{m-1} y^{m-1} b^{m+n}}{b^{m-1} (b+ay)^{m+n}} \frac{ady}{b}, \because \text{limit} \\ &= a^m b^n \int_0^\infty \frac{y^{m-1}}{(ay+b)^{m+n}} dy \end{aligned}$$

Using a property of definite integral, we have

$$\beta(m, n) = a^m b^n \int_0^\infty \frac{x^{m-1}}{(ax+b)^{m+n}} dx, m > 0, n > 0.$$

Deduction. Putting $x = \tan^2 \theta$ in above property 5, we have obtained

$$\int_0^{\frac{\pi}{2}} \frac{\sin^{2m-1} \theta \cdot \cos^{2n-1} \theta}{(a \sin^2 \theta + b \cos^2 \theta)^{m+n}} d\theta = \frac{\beta(m, n)}{2a^m b^n}, m > 0, n > 0.$$

LECTURE NO.:- 20

Properties of Gamma Function**1. Recurrence Formula:** $\Gamma(n+1) = n \Gamma n, n > 0.$ **Proof.** By the definition of the gamma function, we have

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \dots\dots\dots (1)$$

Replacing n by $n+1$ in (1), we get

$$\begin{aligned}\Gamma(n+1) &= \int_0^{\infty} x^{n+1-1} e^{-x} dx \\ &= \int_0^{\infty} x^n e^{-x} dx\end{aligned}$$

Integrating by parts, we have

$$\begin{aligned}\Gamma(n+1) &= \left[x^n \left(\frac{e^{-x}}{-1} \right) \right]_0^{\infty} - \int_0^{\infty} nx^{n-1} \left(\frac{e^{-x}}{-1} \right) dx \\ &= - \left[\lim_{x \rightarrow \infty} \left(\frac{x^n}{e^x} \right) - 0 \right] + n \int_0^{\infty} x^{n-1} e^{-x} dx \\ &= -0 + n \Gamma n, \text{ by (1).}\end{aligned}$$

$$\therefore \boxed{\Gamma(n+1) = n \Gamma n, n > 0.}$$

This is known as functional or factorial relation or reduction or recursion or recurrence formula or difference equation for gamma function.

2. Relation between gamma function and factorial:If n is a non-negative integer, then $\Gamma(n+1) = n!$.**Proof.** By the definition of the gamma function, we have

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \dots\dots\dots (1)$$

Putting $n = 1$ in (1), we get

$$\begin{aligned}\Gamma 1 &= \int_0^{\infty} x^{1-1} e^{-x} dx \\ &= \int_0^{\infty} e^{-x} dx\end{aligned}$$

$$= \left(\frac{e^{-x}}{-1} \right) \Big|_0^{\infty} = -[0 - 1]$$

$$\therefore \boxed{\Gamma 1 = 1.} \dots\dots\dots (2)$$

We know the recurrence relation of gamma function that

$$\Gamma(n+1) = n \Gamma n \dots\dots\dots (3)$$

Putting $n = 1, 2, 3, \dots\dots$ in (3), we get

$$\Gamma 2 = 1 \Gamma 1 = 1, \text{ by (2)} \Rightarrow \Gamma 2 = 1! \dots\dots\dots (4)$$

$$\Gamma 3 = 2 \Gamma 2 = 2 \times 1!, \text{ by (4)} \Rightarrow \Gamma 3 = 2! \dots\dots\dots (5)$$

$$\Gamma 4 = 3 \Gamma 3 = 3 \times 2!, \text{ by (5)} \Rightarrow \Gamma 4 = 3!$$

In general, we have

$$\boxed{\Gamma(n+1) = n!, n \in \mathbb{N}.}$$

Note 1. From above proof, we have

$$\Gamma(n+1) = n(n-1)(n-2) \dots\dots 3 \cdot 2 \cdot 1 = n!, \text{ if } n \in \mathbb{N}.$$

Note 2. Gamma function is regarded as the generalization of the elementary factorial function.

3. Standard Integral: $\int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma n}{a^n}, n > 0, a > 0.$

Proof. By the definition of the gamma function, we have

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \dots\dots\dots (1)$$

Putting $x = az$ in (1), we get

$$\Gamma n = \int_0^{\infty} a^{n-1} z^{n-1} \cdot e^{-az} \cdot adz, n > 0$$

$$= a^n \int_0^{\infty} z^{n-1} e^{-az} dz$$

$$= a^n \int_0^{\infty} x^{n-1} e^{-ax} dx, \text{ by a property of definite integral}$$

$$\therefore \boxed{\int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma n}{a^n}, n > 0, a > 0.}$$

Note. $\Gamma_n = \int_0^\infty x^{n-1} e^{-ax} dx$ is the standard form of the gamma function.

4. Gamma function for zero and negative values (Γ_n for $n \leq 0$) :

Left marching recurrence formula of the gamma function is

$$\Gamma(n+1) = n \Gamma_n$$

$$\Rightarrow \Gamma_n = \frac{\Gamma(n+1)}{n} \dots\dots\dots (1)$$

Putting $n = 0, -1, -2, -3, \dots\dots$ in (1), we get

$$\Gamma_0 = \frac{\Gamma_1}{0} = \frac{1}{0} \rightarrow \pm \infty, \therefore \Gamma_1 = 1 \Rightarrow \Gamma_0 \rightarrow \pm \infty \dots\dots\dots (2)$$

$$\Gamma_{(-1)} = \frac{\Gamma_0}{-1} \rightarrow \pm \infty, \text{ by (2)} \Rightarrow \Gamma_{(-1)} \rightarrow \pm \infty \dots\dots\dots (3)$$

$$\Gamma_{(-2)} = \frac{\Gamma_{(-1)}}{-2} \rightarrow \pm \infty, \text{ by (3)} \Rightarrow \Gamma_{(-2)} \rightarrow \pm \infty \dots\dots\dots (4)$$

$$\Gamma_{(-3)} = \frac{\Gamma_{(-2)}}{-3} \rightarrow \pm \infty, \text{ by (4)} \Rightarrow \Gamma_{(-3)} \rightarrow \pm \infty$$

In general, we have

$$\boxed{\Gamma_n \rightarrow \pm \infty, \text{ for } n = 0, -1, -2, -3, \dots\dots} \dots\dots\dots (5)$$

Hence the gamma function Γ_n is not defined for $n = 0, -1, -2, -3, \dots\dots$.

Now repeated application of (1) gives the result

$$\Gamma_n = \frac{\Gamma(n+1)}{n} = \frac{\Gamma(n+2)}{n(n+1)} = \dots\dots\dots$$

$$\therefore \Gamma_n = \frac{\Gamma(n+k+1)}{n(n+1)\dots\dots(n+k)}.$$

This relation gives the values of gamma function for negative values of n except $n = -1, -2, -3, \dots\dots$.

Note. If n is positive integer, then $\Gamma(n+1) = n!$. So from (5), we have $n! \rightarrow \pm \infty$, for $n = -1, -2, -3, \dots\dots$; i.e. the factorials of negative integers are undefined.

5. Existence and graph of the gamma function

By the definition of the gamma function, we have

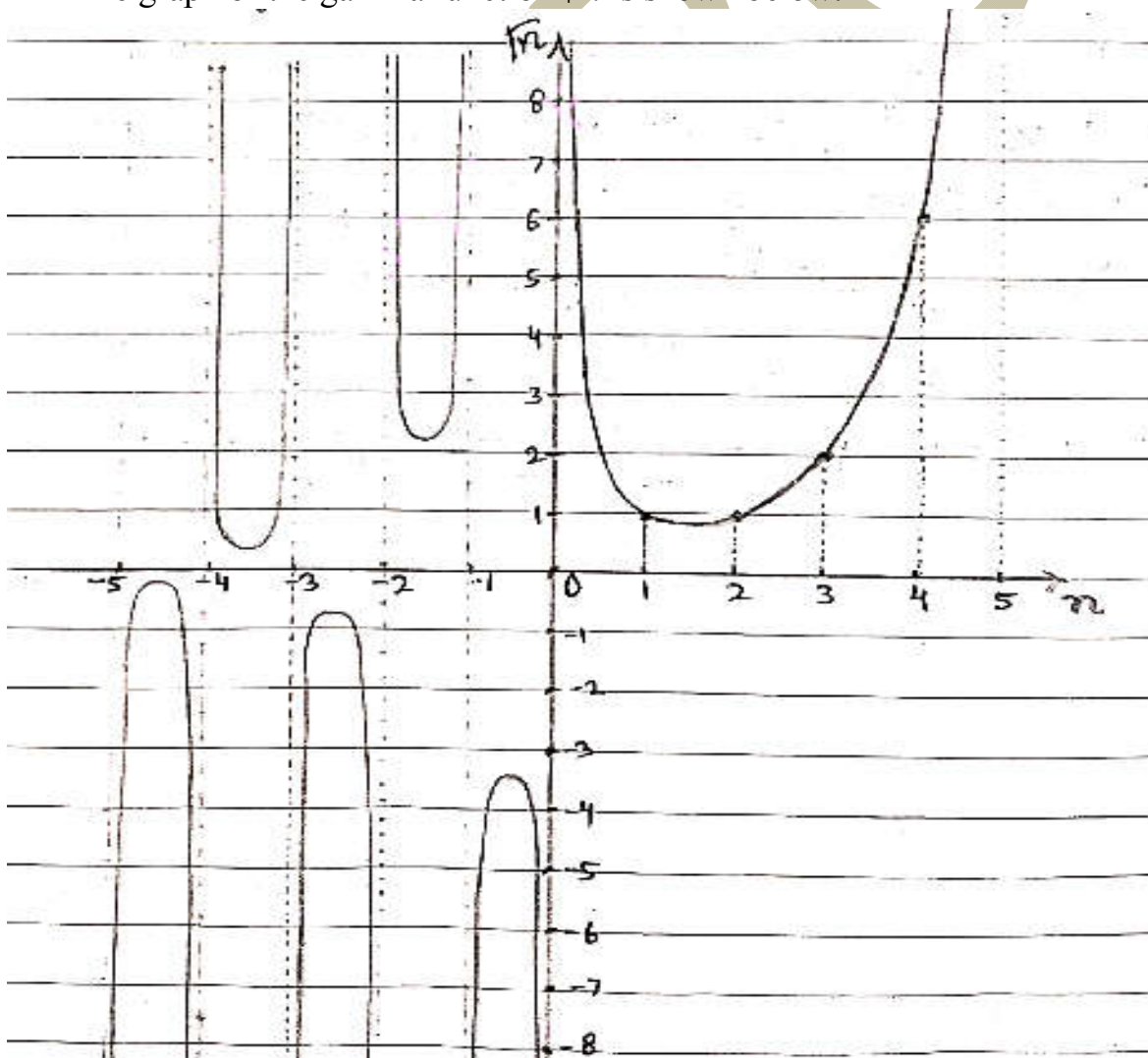
$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0. \dots\dots\dots (1)$$

The gamma function Γn is continuous for $n > 0$. From above property 4, we have concluded that the gamma function Γn is not defined for $n = 0, -1, -2, \dots$ and defined for non-integer negative values of n .

Hence the gamma function Γn is continuous for all real values of n except $n = 0, -1, -2, \dots$; i.e. Γn is discontinuous at $n = 0, -1, -2, \dots$.

Therefore the gamma function Γn is defined for all real values of n except $n = 0, -1, -2, \dots$; i.e. except zero and negative integers.

The graph of the gamma function Γn is shown below:



Some Other Forms of Gamma Function

1. $\Gamma n = \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy, n > 0.$

Proof. By the definition of the gamma function, we have

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \dots\dots\dots (1)$$

Let $x = \log \left(\frac{1}{y}\right)$. So $y = e^{-x}$ and $dy = -e^{-x} dx$ i.e. $e^{-x} dx = -dy$.

Putting these values in (1), we get

$$\Gamma n = \int_1^0 \left(\log \frac{1}{y}\right)^{n-1} (-dy), n > 0, \therefore \text{limit}$$

x	0	∞
$y = e^{-x}$	1	0

$$\therefore \Gamma n = \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy, n > 0, \text{ by a property of definite integral.}$$

2. $\Gamma(n+1) = \int_0^{\infty} e^{-y} y^{\frac{1}{n}} dy = n \Gamma n, n > 0.$

Proof. By the definition of the gamma function, we have

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \dots\dots\dots (1)$$

Let $x = y^{\frac{1}{n}}$. So $x^n = y$ and $nx^{n-1} dx = dy$.

Putting these values in (1), we get

$$\Gamma n = \int_0^{\infty} e^{-y^{\frac{1}{n}}} \frac{dy}{n}, n > 0, \therefore \text{limit}$$

x	0	∞
$y = x^n$	0	∞

$$= \frac{1}{n} \int_0^{\infty} e^{-y^{\frac{1}{n}}} dy, n > 0$$

$$\Rightarrow n \Gamma n = \int_0^{\infty} e^{-y^{\frac{1}{n}}} dy, n > 0.$$

$$\therefore \Gamma(n+1) = n \Gamma n = \int_0^{\infty} e^{-y^{\frac{1}{n}}} dy, n > 0.$$

3. $\Gamma \left(\frac{m+1}{n}\right) = na^{\frac{m+1}{n}} \int_0^{\infty} t^m e^{-at^n} dt, m > 0, n > 0, a > 0.$

Proof. Let $at^n = x$. So $t = \left(\frac{x}{a}\right)^{\frac{1}{n}}$ and $dt = \frac{1}{n} \left(\frac{x}{a}\right)^{\frac{1}{n}-1} \frac{dx}{a}$.

Putting these values in integral $\int_0^\infty t^m e^{-at^n} dt$, we get

$$\begin{aligned} \int_0^\infty t^m e^{-at^n} dt &= \int_0^\infty \left(\frac{x}{a}\right)^{\frac{m}{n}} e^{-x} \frac{1}{an} \left(\frac{x}{a}\right)^{\frac{1}{n}-1} dx, \because \text{limit} \end{aligned}$$

t	0	∞
$x = at^n$	0	∞

$$\begin{aligned} &= \frac{1}{a^{\frac{m}{n}}} \frac{1}{an} a a^{-\frac{1}{n}} \int_0^\infty x^{\frac{m}{n} + \frac{1}{n} - 1} e^{-x} dx \\ &= \frac{1}{n a^{\frac{m}{n} + \frac{1}{n}}} \int_0^\infty x^{\frac{m+1}{n} - 1} e^{-x} dx \\ &= \frac{1}{n a^{\frac{m+1}{n}}} \Gamma\left(\frac{m+1}{n}\right), \text{ by the definition of the gamma function} \end{aligned}$$

$$\therefore \Gamma\left(\frac{m+1}{n}\right) = n a^{\frac{m+1}{n}} \int_0^\infty t^m e^{-at^n} dt, m > 0, n > 0, a > 0.$$

$$4. \Gamma(n+1) = (m+1)^{n+1} (-1)^n \int_0^1 x^m (\log x)^n dx, n > -1, m > -1.$$

Proof. Let $\log x = -y$. So $x = e^{-y}$ and $dx = -e^{-y} dy$.

Putting these values in integral $\int_0^1 x^m (\log x)^n dx$, we get

$$\begin{aligned} \int_0^1 x^m (\log x)^n dx &= \int_\infty^0 e^{-my} (-y)^n (-e^{-y}) dy, \because \text{limit} \end{aligned}$$

x	0	1
$y = -\log x$	∞	0

$$\begin{aligned} &= \int_0^\infty (-1)^n y^n e^{-my-y} dy, \text{ by a property of definite integral} \\ &= (-1)^n \int_0^\infty y^{(n+1)-1} e^{-(m+1)y} dy \\ &= (-1)^n \frac{\Gamma(n+1)}{(m+1)^{n+1}}, n > -1, m > -1; \end{aligned}$$

$$\text{by } \int_0^\infty y^{n-1} e^{-ay} dy = \frac{\Gamma n}{a^n}, n > 0, a > 0.$$

$$\Rightarrow (-1)^n \Gamma(n+1) = (m+1)^{n+1} \int_0^1 x^m (\log x)^n dx$$

Multiplying both sides by $(-1)^n$, we get

$$\Gamma(n+1) = (m+1)^{n+1}(-1)^n \int_0^1 x^m (\log x)^n dx, n > -1, m > -1.$$

$$5. \int_0^1 x^{n-1} \left(\log \frac{1}{x}\right)^{m-1} dx = \frac{\Gamma m}{n^m}, m > 0, n > 0.$$

Proof. Let $\log \frac{1}{x} = y$. So $x = e^{-y}$ and $dx = -e^{-y} dy$.

Putting these values in integral $\int_0^1 x^{n-1} \left(\log \frac{1}{x}\right)^{m-1} dx$, we get

$$\begin{aligned} \int_0^1 x^{n-1} \left(\log \frac{1}{x}\right)^{m-1} dx &= \int_{\infty}^0 (e^{-y})^{n-1} y^{m-1} (-e^{-y}) dy, \because \text{limit} \\ &= \int_0^{\infty} e^{-y(n-1)-y} y^{m-1} dy, \text{ by a property of definite integral} \\ &= \int_0^{\infty} y^{m-1} e^{-ny} dy \\ &= \frac{\Gamma m}{n^m}, m > 0, n > 0; \text{ by } \int_0^{\infty} y^{n-1} e^{-ay} dy = \frac{\Gamma n}{a^n}, n > 0, a > 0. \end{aligned}$$

$$\therefore \int_0^1 x^{n-1} \left(\log \frac{1}{x}\right)^{m-1} dx = \frac{\Gamma m}{n^m}, m > 0, n > 0.$$

LECTURE NO.:- 21

Relation between Beta and Gamma Functions:

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}, m > 0, n > 0.$$

Proof. We know that

$$\int_0^\infty x^{n-1} e^{-zx} dx = \frac{\Gamma(n)}{z^n}, n > 0, z > 0. \dots\dots\dots (1)$$

$$\Rightarrow \Gamma(n) = z^n \int_0^\infty x^{n-1} e^{-zx} dx$$

$$\Rightarrow \Gamma(n) = \int_0^\infty z^n x^{n-1} e^{-zx} dx. \dots\dots\dots (2)$$

Multiplying both sides of (2) by $e^{-z} z^{m-1}$ and integrating w.r.t. z from 0 to ∞ , we get

$$\int_0^\infty \Gamma(n) \cdot e^{-z} z^{m-1} dz = \int_0^\infty \{e^{-z} z^{m-1} \int_0^\infty z^n e^{-zx} x^{n-1} dx\} dz$$

$$\Rightarrow \Gamma(n) \cdot \int_0^\infty e^{-z} z^{m-1} dz = \int_0^\infty \left\{ \int_0^\infty e^{-z} z^{m-1} z^n e^{-zx} x^{n-1} dx \right\} dz. \dots\dots\dots (3)$$

Using the definition of the gamma function in L.H.S. of (3), we get

$$\Gamma(n) \cdot \Gamma(m) = \int_0^\infty \left\{ \int_0^\infty e^{-z(1+x)} z^{m-1+n} x^{n-1} dx \right\} dz$$

$$\Rightarrow \Gamma(m) \cdot \Gamma(n) = \int_0^\infty \left\{ \int_0^\infty z^{m+n-1} e^{-(1+x)z} x^{n-1} dx \right\} dz. \dots\dots\dots (4)$$

Changing the order of integration in (4), we have

$$\Gamma(m) \cdot \Gamma(n) = \int_0^\infty \left\{ \int_0^\infty z^{m+n-1} e^{-(1+x)z} x^{n-1} dz \right\} dx$$

$$= \int_0^\infty \left\{ x^{n-1} \int_0^\infty z^{m+n-1} e^{-(1+x)z} dz \right\} dx$$

$$= \int_0^\infty \left\{ x^{n-1} \frac{\Gamma(m+n)}{(1+x)^{m+n}} \right\} dx, \text{ by (1)}$$

$$= \Gamma(m+n) \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$= \Gamma(m+n) \cdot \beta(m, n), \text{ by a form of beta function.}$$

$$\therefore \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}, m > 0, n > 0.$$

Corollary: Using above relation in $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0$, we get

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0.$$

Deductions/ Formulae of Beta and Gamma Functions

1. Gamma Formula: $\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$

Proof. We know that

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0$$

Putting $2m - 1 = p$ and $2n - 1 = q$ i.e. $m = \frac{p+1}{2}$ and $n = \frac{q+1}{2}$, we get

$$\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)} = 2 \int_0^{\frac{\pi}{2}} \sin^p \theta \cdot \cos^q \theta d\theta$$

Replacing p by m and q by n , we get

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

2. $\beta(m, n) = \beta(m+1, n) + \beta(m, n+1).$

Proof. We know that

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \dots \dots \dots (1)$$

Replacing m by $m+1$ in (1), we get

$$\beta(m+1, n) = \frac{\Gamma(m+1) \Gamma(n)}{\Gamma(m+1+n)} \dots \dots \dots (2)$$

Replacing n by $n+1$ in (1), we get

$$\beta(m, n+1) = \frac{\Gamma(m) \Gamma(n+1)}{\Gamma(m+n+1)} \dots \dots \dots (3)$$

Adding (2) and (3), we get

$$\begin{aligned}
 \beta(m+1, n) + \beta(m, n+1) &= \frac{\Gamma(m+1) \Gamma n}{\Gamma(m+1+n)} + \frac{\Gamma m \Gamma(n+1)}{\Gamma(m+n+1)} \\
 &= \frac{\Gamma(m+1) \Gamma n + \Gamma m \Gamma(n+1)}{\Gamma(m+n+1)} \\
 &= \frac{m \Gamma m \Gamma n + \Gamma m \cdot n \Gamma n}{(m+n) \Gamma(m+n)}, \text{ by } \Gamma(n+1) = n \Gamma n \\
 &= \frac{\Gamma m \Gamma n (m+n)}{(m+n) \Gamma(m+n)} \\
 &= \frac{\Gamma m \Gamma n}{\Gamma(m+n)} \\
 &= \beta(m, n), \text{ by (1)}
 \end{aligned}$$

$$\therefore \beta(m, n) = \beta(m+1, n) + \beta(m, n+1).$$

$$3. \Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi} = \pi \operatorname{cosec} n\pi.$$

Proof. We know that

$$\begin{aligned}
 \beta(m, n) &= \frac{\Gamma m \Gamma n}{\Gamma(m+n)} \text{ \& } \beta(m, n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0 \\
 \Rightarrow \frac{\Gamma m \Gamma n}{\Gamma(m+n)} &= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx, m > 0, n > 0. \\
 \text{Putting } m+n &= 1 \text{ i.e. } m = 1-n, \text{ we get} \\
 \frac{\Gamma(1-n) \Gamma n}{\Gamma 1} &= \int_0^\infty \frac{x^{n-1}}{(1+x)^1} dx, n > 0 \\
 \Rightarrow \frac{\Gamma n \Gamma(1-n)}{1} &= \int_0^\infty \frac{x^{n-1}}{1+x} dx, n > 0, \because \Gamma 1 = 1. \dots\dots\dots (1)
 \end{aligned}$$

$$\text{A standard integral is } \int_0^\infty \frac{x^{n-1}}{1+x} dx = \frac{\pi}{\sin n\pi}. \dots\dots\dots (2)$$

From (1) and (2), we have

$$\boxed{\Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi} = \pi \operatorname{cosec} n\pi.}$$

4. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof. We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Putting $m = 0 = n$, we get

$$\int_0^{\frac{\pi}{2}} 1 \times 1 d\theta = \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{2 \Gamma\left(\frac{2}{2}\right)}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} d\theta = \frac{\left\{\Gamma\left(\frac{1}{2}\right)\right\}^2}{2 \Gamma 1} \Rightarrow [\theta]_0^{\frac{\pi}{2}} = \frac{\left\{\Gamma\left(\frac{1}{2}\right)\right\}^2}{2 \times 1}, \because \Gamma 1 = 1$$

$$\Rightarrow \left[\frac{\pi}{2} - 0\right] = \frac{1}{2} \left\{\Gamma\left(\frac{1}{2}\right)\right\}^2 \Rightarrow \pi = \left\{\Gamma\left(\frac{1}{2}\right)\right\}^2.$$

$$\therefore \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}.$$

5. Γn for $n = \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \dots$.

We know the recurrence relation of the gamma function that

$$\Gamma(n+1) = n \Gamma n, n > 0. \dots\dots\dots (1)$$

Putting $n = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$ in (1), we get

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{1}{2} \sqrt{\pi}, \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}, \because \Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \sqrt{\pi}$$

$$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \Gamma\left(\frac{5}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}, \because \Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}$$

.....

Hence generally, we have

$$\Gamma\left(\frac{2n+1}{2}\right) = \left(\frac{2n-1}{2}\right) \left(\frac{2n-3}{2}\right) \left(\frac{2n-5}{2}\right) \dots\dots \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}.$$

6. Γn for $n = -\frac{1}{2}, -\frac{3}{2}, -\frac{5}{2}, \dots$.

We know the recurrence relation of the gamma function that

$$\Gamma(n+1) = n \Gamma n$$

$$\Rightarrow \Gamma n = \frac{1}{n} \Gamma(n+1). \dots\dots\dots (1)$$

Putting $n = -\frac{1}{2}, -\frac{3}{2}, -\frac{5}{2}, \dots$ in (1), we get

$$\Gamma\left(-\frac{1}{2}\right) = -2 \Gamma\left(\frac{1}{2}\right) = -2\sqrt{\pi}, \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(-\frac{3}{2}\right) = -\frac{2}{3} \Gamma\left(-\frac{1}{2}\right) = \left(-\frac{2}{3}\right)(-2)\sqrt{\pi} = \frac{4}{3}\sqrt{\pi}, \because \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

$$\Gamma\left(-\frac{5}{2}\right) = -\frac{2}{5} \Gamma\left(-\frac{3}{2}\right) = \left(-\frac{2}{5}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{1}\right)\sqrt{\pi} = -\frac{8}{15}\sqrt{\pi}$$

.....

Hence generally, we have

$$\Gamma\left(-\frac{2n+1}{2}\right) = \left(-\frac{2}{2n+1}\right)\left(-\frac{2}{2n-1}\right)\left(-\frac{2}{2n-3}\right)\dots\dots\left(-\frac{2}{5}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{1}\right)\sqrt{\pi}.$$

7. Legendre's Duplication Formula: $\Gamma m \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m), m > 0.$

Proof. We know that

$$\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)} = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0$$

$$\Rightarrow \frac{\Gamma m \Gamma n}{2 \Gamma(m+n)} = \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta, m > 0, n > 0. \dots\dots\dots (1)$$

Putting $n = \frac{1}{2}$ in (1), we get

$$\frac{\Gamma m \Gamma\left(\frac{1}{2}\right)}{2 \Gamma\left(m + \frac{1}{2}\right)} = \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta d\theta, m > 0. \dots\dots\dots (2)$$

Again putting $n = m$ in (1), we get

$$\begin{aligned}
 \frac{(\Gamma m)^2}{2 \Gamma(2m)} &= \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cdot \cos^{2m-1} \theta d\theta, m > 0 \\
 &= \int_0^{\frac{\pi}{2}} \left(\frac{2}{2} \sin \theta \cdot \cos \theta \right)^{2m-1} d\theta \\
 \Rightarrow \frac{(\Gamma m)^2}{2 \Gamma(2m)} &= \frac{1}{2^{2m-1}} \int_0^{\frac{\pi}{2}} (\sin 2\theta)^{2m-1} d\theta \dots\dots\dots (3)
 \end{aligned}$$

Putting $2\theta = \varphi$ in (3), we get

$$\begin{aligned}
 \frac{(\Gamma m)^2}{2 \Gamma(2m)} &= \frac{1}{2^{2m-1}} \int_0^{\pi} (\sin \varphi)^{2m-1} \frac{d\varphi}{2} \\
 &= \frac{1}{2^{2m-1}} \frac{1}{2} \int_0^{\pi} \sin^{2m-1} \varphi d\varphi \\
 &= \frac{1}{2^{2m-1}} \frac{1}{2} 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \varphi d\varphi, \because \sin^{2m-1}(\pi - \varphi) = \sin^{2m-1} \varphi \\
 &= \frac{1}{2^{2m-1}} \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta d\theta, \text{ by a property of definite integral} \\
 \Rightarrow \int_0^{\pi} \sin^{2m-1} \theta d\theta &= \frac{2^{2m-1} (\Gamma m)^2}{2 \Gamma(2m)}, m > 0.
 \end{aligned}$$

Putting this value in (2), we get

$$\begin{aligned}
 \frac{\Gamma m \Gamma\left(\frac{1}{2}\right)}{2 \Gamma\left(m + \frac{1}{2}\right)} &= \frac{2^{2m-1} (\Gamma m)^2}{2 \Gamma(2m)}, m > 0 \\
 \Rightarrow \frac{\sqrt{\pi}}{\Gamma\left(m + \frac{1}{2}\right)} &= \frac{2^{2m-1} \Gamma m}{\Gamma(2m)}, \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.
 \end{aligned}$$

$$\therefore \Gamma m \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m), m > 0.$$

LECTURE NO.:- 22

Problems on Beta and Gamma Functions

1. Evaluate $\beta\left(\frac{5}{2}, \frac{3}{2}\right)$.

Solution. We know that

$$\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}, m > 0, n > 0.$$

Putting $m = \frac{5}{2}$ and $n = \frac{3}{2}$, we get

$$\begin{aligned}\beta\left(\frac{5}{2}, \frac{3}{2}\right) &= \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{5}{2} + \frac{3}{2}\right)} = \frac{\left(\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right) \left(\frac{1}{2} \sqrt{\pi}\right)}{\Gamma 4}, \because \Gamma(n+1) = n \Gamma n \text{ \& } \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \\ &= \frac{3\pi}{8 \times 3!} = \frac{3\pi}{8 \times 3 \times 2}.\end{aligned}$$

$$\therefore \beta\left(\frac{5}{2}, \frac{3}{2}\right) = \frac{\pi}{16}. \quad \underline{\text{Ans.}}$$

2. Evaluate $\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$.

Solution. We know that

$$\Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi}.$$

Putting $n = \frac{1}{4}$, we get

$$\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{\pi}{\sin\left(\frac{\pi}{4}\right)} = \frac{\pi}{1/\sqrt{2}}$$

$$\therefore \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \pi \sqrt{2}. \quad \underline{\text{Ans.}}$$

3. Evaluate $\int_0^{\frac{\pi}{2}} \sin^6 \theta \cdot \cos^7 \theta d\theta$.

Solution. We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Putting $m = 6$ and $n = 7$, we get

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^6 \theta \cdot \cos^7 \theta d\theta &= \frac{\Gamma\left(\frac{6+1}{2}\right) \Gamma\left(\frac{7+1}{2}\right)}{2 \Gamma\left(\frac{6+7+2}{2}\right)} = \frac{\Gamma\left(\frac{7}{2}\right) \Gamma 4}{2 \Gamma\left(\frac{15}{2}\right)} \\ &= \frac{\Gamma\left(\frac{7}{2}\right) \times 3!}{2 \times \frac{13}{2} \cdot \frac{11}{2} \cdot \frac{9}{2} \cdot \frac{7}{2} \Gamma\left(\frac{7}{2}\right)}, \text{ by } \Gamma(n+1) = n \Gamma n \\ &= \frac{3 \times 2}{13 \cdot \frac{11}{2} \cdot \frac{9}{2} \cdot \frac{7}{2}} = \frac{3 \times 2 \times 8}{13 \times 11 \times 3 \times 3 \times 7} = \frac{16}{13 \times 11 \times 3 \times 7}.\end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin^6 \theta \cdot \cos^7 \theta d\theta = \frac{16}{3003}. \quad \underline{\text{Ans.}}$$

4. Evaluate $\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta$.

Solution. We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Putting $m = 10$ and $n = 0$, we get

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta &= \frac{\Gamma\left(\frac{10+1}{2}\right) \Gamma\left(\frac{0+1}{2}\right)}{2 \Gamma\left(\frac{10+0+2}{2}\right)} = \frac{\Gamma\left(\frac{11}{2}\right) \Gamma\left(\frac{1}{2}\right)}{2 \Gamma 6} \\ &= \frac{\frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \sqrt{\pi}}{2 \times 5!} = \frac{9 \cdot 7 \cdot 5 \cdot 3 \pi}{2^6 \times 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= \frac{9 \times 7 \pi}{2^6 \times 4 \times 2} = \frac{63 \pi}{2^9}.\end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta = \frac{63\pi}{512}. \quad \underline{\text{Ans.}}$$

5. Evaluate $\int_0^{\frac{\pi}{2}} \cos^9 \theta d\theta$.

Solution. We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Putting $m = 0$ and $n = 9$, we get

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \cos^9 \theta \, d\theta &= \frac{\Gamma\left(\frac{0+1}{2}\right) \Gamma\left(\frac{9+1}{2}\right)}{2 \Gamma\left(\frac{0+9+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{1}{2}\right) \Gamma 5}{2 \Gamma\left(\frac{11}{2}\right)} = \frac{\sqrt{\pi} 4!}{2 \times \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}} \\
 &= \frac{4 \cdot 3 \cdot 2 \cdot 1 \times 16}{9 \cdot 7 \cdot 5 \cdot 3} = \frac{8 \times 16}{9 \cdot 7 \cdot 5}.
 \end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{2}} \cos^9 \theta \, d\theta = \frac{128}{315}. \quad \underline{\text{Ans.}}$$

6. Evaluate $\int_0^{\frac{\pi}{2}} \left(\frac{\sqrt[3]{\sin^8 x}}{\sqrt{\cos x}} \right) dx$.

Solution. We have

$$\int_0^{\frac{\pi}{2}} \left(\frac{\sqrt[3]{\sin^8 x}}{\sqrt{\cos x}} \right) dx = \int_0^{\frac{\pi}{2}} \sin^{\frac{8}{3}} x \cdot \cos^{-\frac{1}{2}} x \, dx. \quad \dots\dots\dots (1)$$

We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta \, d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}. \quad \dots\dots\dots (2)$$

Putting $m = \frac{8}{3}$, $n = -\frac{1}{2}$ and $\theta = x$ in (2), we get

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^{\frac{8}{3}} x \cdot \cos^{-\frac{1}{2}} x \, dx &= \frac{\Gamma\left(\frac{\frac{8}{3}+1}{2}\right) \Gamma\left(\frac{-\frac{1}{2}+1}{2}\right)}{2 \Gamma\left(\frac{\frac{8}{3}-\frac{1}{2}+2}{2}\right)} = \frac{\Gamma\left(\frac{11}{6}\right) \Gamma\left(\frac{1}{4}\right)}{2 \Gamma\left(\frac{25}{12}\right)} \\
 &= \frac{\frac{5}{6} \Gamma\left(\frac{5}{6}\right) \times \Gamma\left(\frac{1}{4}\right)}{2 \times \frac{13}{12} \cdot \frac{1}{12} \Gamma\left(\frac{1}{12}\right)}, \text{ by } \Gamma(n+1) = n \Gamma n \\
 &= \frac{60 \Gamma\left(\frac{5}{6}\right) \times \Gamma\left(\frac{1}{4}\right)}{13 \Gamma\left(\frac{1}{12}\right)}.
 \end{aligned}$$

Putting it in (1), we get

$$\therefore \int_0^{\frac{\pi}{2}} \left(\frac{\sqrt[3]{\sin^8 x}}{\sqrt{\cos x}} \right) dx = \frac{60 \Gamma\left(\frac{5}{6}\right) \times \Gamma\left(\frac{1}{4}\right)}{13 \Gamma\left(\frac{1}{12}\right)}. \quad \underline{\text{Ans.}}$$

7. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$.

Solution. We have

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta = \int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}} \theta \cdot \cos^{-\frac{1}{2}} \theta d\theta \dots\dots\dots (1)$$

We know that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)} \dots\dots\dots (2)$$

Putting $m = \frac{1}{2}$ and $n = -\frac{1}{2}$ in (2), we get

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}} \theta \cdot \cos^{-\frac{1}{2}} \theta d\theta &= \frac{\Gamma\left(\frac{\frac{1}{2}+1}{2}\right) \Gamma\left(\frac{-\frac{1}{2}+1}{2}\right)}{2 \Gamma\left(\frac{\frac{1}{2}-\frac{1}{2}+2}{2}\right)} \\ &= \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{2 \Gamma 1} = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right) \\ &= \frac{1}{2} \frac{\pi}{\sin \frac{1}{4}\pi}, \text{ by } \Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi} \\ &= \frac{\pi}{2\left(\frac{1}{\sqrt{2}}\right)}. \end{aligned}$$

Putting it in (1), we get

$$\therefore \int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta = \frac{\pi}{\sqrt{2}}. \text{ Ans.}$$

8. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\cot \theta} d\theta$.

Solution. It is same as above example 7.

9. Evaluate $\int_0^{\infty} \frac{x^2 dx}{(1+x^4)^3}$.

Solution. Let $x^4 = \tan^2 \theta$ so that $x = \sqrt{\tan \theta}$ and $dx = \frac{1}{2} (\tan \theta)^{-\frac{1}{2}} \sec^2 \theta d\theta$.

Putting these values in integral $\int_0^{\infty} \frac{x^2 dx}{(1+x^4)^3}$, we get

$$\int_0^{\infty} \frac{x^2 dx}{(1+x^4)^3} = \int_0^{\frac{\pi}{2}} \frac{\tan \theta \cdot \frac{1}{2} (\tan \theta)^{-\frac{1}{2}} \sec^2 \theta d\theta}{(\sec^2 \theta)^3}, \because \text{limit}$$

x	0	∞
$\tan \theta = x^2$	0	∞
θ	0	$\pi/2$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{(\tan \theta)^{\frac{1}{2}}}{\sec^4 \theta} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{(\sin \theta)^{\frac{1}{2}} \cos^4 \theta}{(\cos \theta)^{\frac{1}{2}}} d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}} \theta \cdot \cos^{\frac{7}{2}} \theta d\theta \\
 &= \frac{1}{2} \frac{\Gamma\left(\frac{\frac{1}{2}+1}{2}\right) \Gamma\left(\frac{\frac{7}{2}+1}{2}\right)}{2 \Gamma\left(\frac{\frac{1}{2}+\frac{7}{2}+2}{2}\right)}, \text{ by gamma function formula} \\
 &= \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{9}{4}\right)}{4 \Gamma 3} = \frac{\Gamma\left(\frac{3}{4}\right) \times \frac{5}{4} \Gamma\left(\frac{1}{4}\right)}{4 \times 2!}, \text{ by } \Gamma(n+1) = n \Gamma n \\
 &= \frac{5 \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)}{16 \times 8} = \frac{5}{128} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right) \\
 &= \frac{5}{128} \frac{\pi}{\sin \frac{1}{4} \pi}, \text{ by } \Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi} \\
 &= \frac{5\pi}{128 \left(\frac{1}{\sqrt{2}}\right)}.
 \end{aligned}$$

$$\therefore \int_0^{\infty} \frac{x^2 dx}{(1+x^4)^3} = \frac{5\pi\sqrt{2}}{128}. \quad \underline{\text{Ans.}}$$

10. Evaluate $\int_0^{\infty} \frac{dx}{1+x^4}$.

Solution. It is same as above example 9.

11. Evaluate $\int_0^{\infty} \frac{x dx}{1+x^6}$.

Solution. It is same as above example 9.

12. Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 6\theta \cdot \cos^4 3\theta d\theta$.

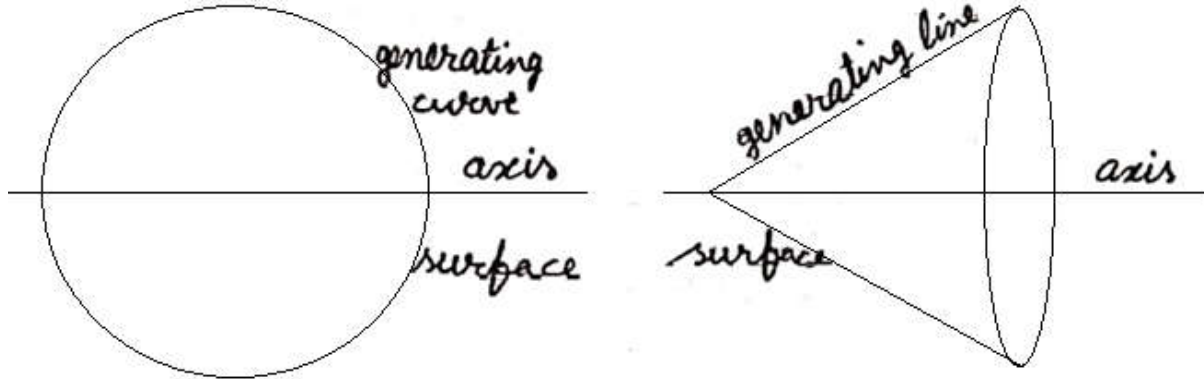
Hint. Let $3\theta = x$ and use gamma function formula.

LECTURE NO.:- 23

Surface Areas of Solids of Revolution

When an arc of a plane curve is revolved about a fixed line lying on the plane of the curve, a surface is generated, which is called the surface of revolution. The fixed line is called the axis of revolution or rotation. The rotating curve is called the generator of the surface.

For example, when a semi circle revolves about its bounding diameter, it generates the surface of a spherical shell; when a straight line revolves about a coplanar axis, it generates a conical surface.



When a closed curve is rotated about a straight line lying on the plane of the curve, the space bounded by the surface of revolution is called the solid of revolution. A solid of revolution may be formed by the surface of revolution of an open curve and the plane(s) perpendicular to the axis of revolution.

Here following cases may be:

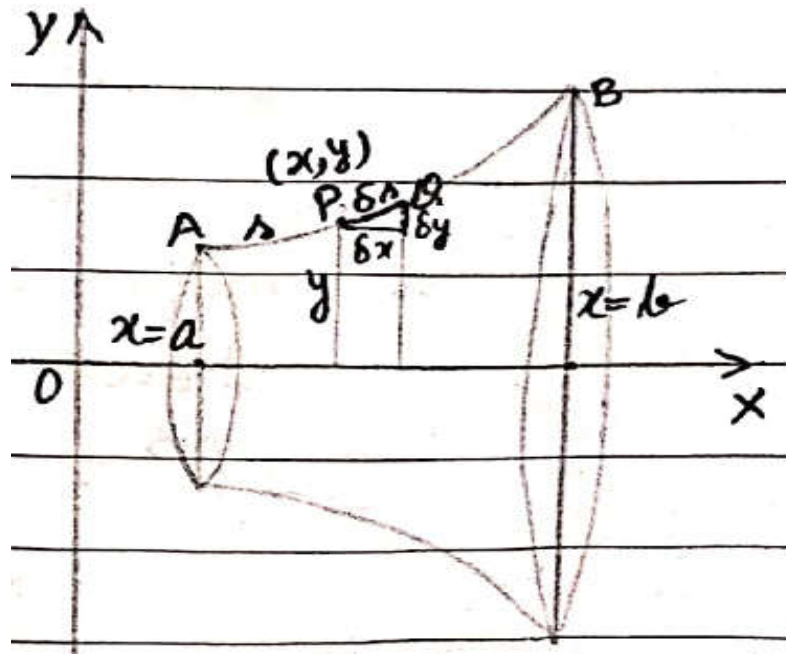
1. When the curve is given in cartesian form $y = f(x)$ or $x = F(y)$:

(i) Revolution about x-axis:

The area of the surface of the solid generated by the revolution of the arc of the curve $y = f(x)$ between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ about x-axis is given by

$$S = \int_A^B 2\pi y \, ds = 2\pi \int_{x_1}^{x_2} f(x) \frac{ds}{dx} dx, \because \text{elementary area} = 2\pi y ds,$$

where s is the length of the arc of the curve from the point A to any point P(x, y).



Here $ds^2 = dx^2 + dy^2 \Rightarrow \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \{f'(x)\}^2}$.

Alternatively, if the equation of the curve is $x = F(y)$, then

$$S = \int_A^B 2\pi y \, ds = 2\pi \int_{y_1}^{y_2} y \frac{ds}{dy} dy, \text{ where } \frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + \{F'(y)\}^2}.$$

(ii) Revolution about y-axis:

The area of the surface of the solid generated by the revolution of the arc of the curve $y = f(x)$ between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ about y-axis is given by

$$S = \int_A^B 2\pi x \, ds = 2\pi \int_{x_1}^{x_2} x \frac{ds}{dx} dx, \text{ where } s \text{ and } \frac{ds}{dx} \text{ are same as in (i).}$$

Alternatively, if the equation of the curve is $x = F(y)$, then

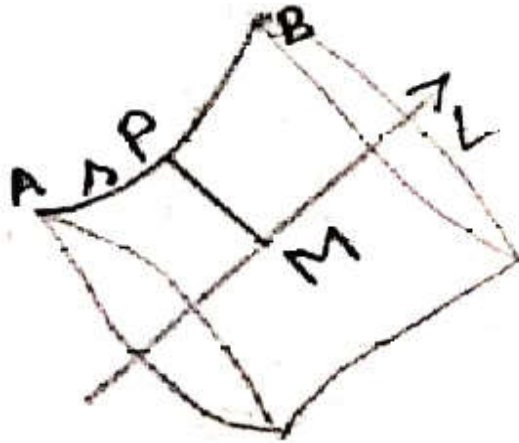
$$S = \int_A^B 2\pi x \, ds = 2\pi \int_{y_1}^{y_2} F(y) \frac{ds}{dy} dy, \text{ where } s \text{ and } \frac{ds}{dy} \text{ are same as in (i).}$$

(iii) Revolution about any line:

The area of the surface of the solid generated by the revolution of the arc of the curve $y = f(x)$ between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ about any given line L is given by

$$S = \int_A^B 2\pi PM \, ds = 2\pi \int_{x_1}^{x_2} PM \frac{ds}{dx} dx,$$

where P is any point (x, y) on the arc AB of the curve, PM is the length of the perpendicular on line L from the point P, and s & $\frac{ds}{dx}$ are same as in (i).



Alternatively, if the equation of the curve is $x = F(y)$, then

$$S = \int_A^B 2\pi PM \, ds = 2\pi \int_{y_1}^{y_2} PM \frac{ds}{dy} dy, \text{ where } \frac{ds}{dy} \text{ is same as in (i).}$$

2. When the curve is given in parametric form $x = x(t)$, $y = y(t)$:

(i) Revolution about x-axis:

The area of the surface of the solid generated by the revolution of the arc of the curve $x = x(t)$, $y = y(t)$ between the points $A(t_1)$ and $B(t_2)$ about x-axis is given by

$$S = \int_A^B 2\pi y \, ds = 2\pi \int_{t_1}^{t_2} y(t) \frac{ds}{dt} dt,$$

where s is the length of the arc of the curve from the point A to any point P(t).

$$\text{Here } ds^2 = dx^2 + dy^2 \Rightarrow \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{\{x'(t)\}^2 + \{y'(t)\}^2}.$$

(ii) Revolution about y-axis:

The area of the surface of the solid generated by the revolution of the arc of the curve $x = x(t)$, $y = y(t)$ between the points $A(t_1)$ and $B(t_2)$ about y-axis is given by

$$S = \int_A^B 2\pi x \, ds = 2\pi \int_{t_1}^{t_2} x(t) \frac{ds}{dt} dt, \text{ where } s \text{ and } \frac{ds}{dt} \text{ are same as in (i).}$$

(iii) Revolution about any line:

The area of the surface of the solid generated by the revolution of the arc of the curve $x = x(t)$, $y = y(t)$ between the points $A(t_1)$ and $B(t_2)$ about any given line L is given by

$$S = \int_A^B 2\pi PM \, ds = 2\pi \int_{t_1}^{t_2} PM \frac{ds}{dt} dt,$$

where $P(t)$ is any point on the arc AB of the curve, PM is the length of the perpendicular on line L from the point P , and s & $\frac{ds}{dt}$ are same as in (i).

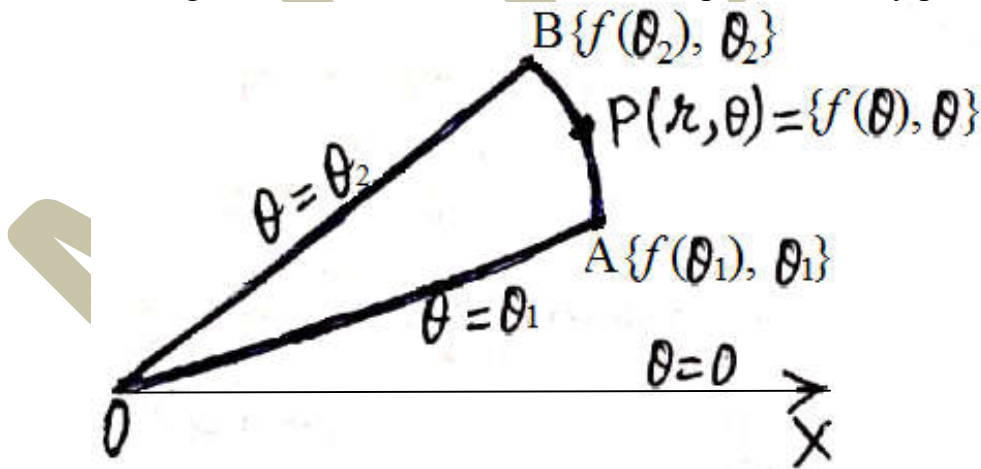
3. When the curve is given in polar form $r = f(\theta)$ or $\theta = F(r)$:

(i) Revolution about initial line $\theta = 0$ (x-axis):

The area of the surface of the solid generated by the revolution of the arc of the curve $r = f(\theta)$ between the points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ about initial line $\theta = 0$ (x-axis) is given by

$$S = \int_A^B 2\pi y \, ds = 2\pi \int_{\theta_1}^{\theta_2} r \sin \theta \frac{ds}{d\theta} d\theta,$$

where s is the length of the arc of the curve from the point A to any point $P(r, \theta)$.



$$\text{Here } ds^2 = dr^2 + (rd\theta)^2 \Rightarrow \frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \sqrt{\{f(\theta)\}^2 + \{f'(\theta)\}^2}.$$

Alternatively, if the equation of the curve is $\theta = F(r)$, then

$$S = 2\pi \int_{r_1}^{r_2} r \sin \theta \frac{ds}{dr} dr, \text{ where } \frac{ds}{dr} = \sqrt{1 + r^2 \left(\frac{d\theta}{dr}\right)^2} = \sqrt{1 + r^2 \{F'(r)\}^2}.$$

(ii) Revolution about line through the pole perpendicular to the initial line i.e. about line $\theta = \pi/2$ (y-axis):

The area of the surface of the solid generated by the revolution of the arc of the curve $r = f(\theta)$ between the points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ about line $\theta = \pi/2$ (y-axis) is given by

$$S = \int_A^B 2\pi x \, ds = 2\pi \int_{\theta_1}^{\theta_2} r \cos \theta \frac{ds}{d\theta} d\theta,$$

where s and $\frac{ds}{d\theta}$ are same as in (i).

Alternatively, if the equation of the curve is $\theta = F(r)$, then

$$S = 2\pi \int_{r_1}^{r_2} r \cos \theta \frac{ds}{dr} dr, \text{ where } s \text{ and } \frac{ds}{dr} \text{ are same as in (i).}$$

(iii) Revolution about any line:

The area of the surface of revolution of the arc of the curve $r = f(\theta)$ between the points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ about any given line L is given by

$$S = \int_A^B 2\pi PM \, ds = 2\pi \int_{\theta_1}^{\theta_2} PM \frac{ds}{d\theta} d\theta,$$

where $P(r, \theta)$ is any point on the arc AB of the curve, PM is the length of the perpendicular on line L from the point P and s & $\frac{ds}{d\theta}$ are same as in (i).

Alternatively, if the equation of the curve is $\theta = F(r)$, then

$$S = \int_A^B 2\pi PM \, ds = 2\pi \int_{r_1}^{r_2} PM \frac{ds}{dr} dr, \text{ where } \frac{ds}{dr} \text{ is same as in (i).}$$

LECTURE NO.:- 24

Problems on Surfaces of Solids of Revolution

1. The arc of the parabola $y^2 = 4x$ between $x = \frac{1}{2}$ and $x = 1$ is rotated about the x -axis. Find the area of the surface generated.

Solution. Given curve is parabola $y^2 = 4x$

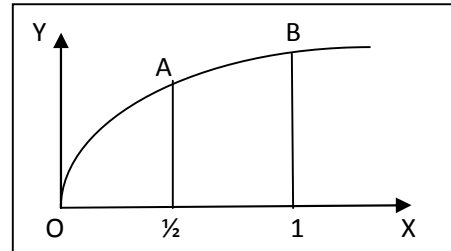
$\Rightarrow y = 2\sqrt{x}$ for the portion in first quadrant.

Differentiating w.r.t. x , $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$.

Here y is positive for $\frac{1}{2} \leq x \leq 1$ and differentiable in this interval.

$\therefore$ The area of the surface generated is

$$\begin{aligned}
 S &= \int_A^B 2\pi y \, ds = 2\pi \int_{x_1}^{x_2} y \frac{ds}{dx} dx \\
 &= 2\pi \int_{\frac{1}{2}}^1 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_{\frac{1}{2}}^1 2\sqrt{x} \sqrt{1 + \frac{1}{x}} dx \\
 &= 2\pi \int_{\frac{1}{2}}^1 \sqrt{x} \frac{\sqrt{x+1}}{\sqrt{x}} dx = 4\pi \int_{\frac{1}{2}}^1 \sqrt{x+1} dx \\
 &= 4\pi \left[\frac{(x+1)^{3/2}}{3/2} \right]_{\frac{1}{2}}^1 = 4\pi \left[\frac{2}{3} (2\sqrt{2}) - \frac{2}{3} \left(\frac{3}{2} \sqrt{\frac{3}{2}} \right) \right] \\
 &= 4\pi \left[\frac{4\sqrt{2}}{3} - \frac{\sqrt{3}\sqrt{2}}{\sqrt{2}\sqrt{2}} \right] = 4\pi \left[\frac{4\sqrt{2}}{3} - \frac{\sqrt{6}}{2} \right] \\
 &= 4\pi \left[\frac{2 \times 4\sqrt{2} - 3\sqrt{6}}{3 \times 2} \right] = 2\pi \left[\frac{8\sqrt{2} - 3\sqrt{6}}{3} \right] \\
 &= \frac{2\pi}{3} (8\sqrt{2} - 3\sqrt{6}). \quad \text{Ans.}
 \end{aligned}$$



2. The portion of the curve $y = \log x$ for $1 \leq x \leq e$ is rotated about y -axis. Find the area of the surface generated.

Solution. Given that the portion of the curve $y = \log x$ for $1 \leq x \leq e$ is rotated about y -axis. Differentiating it w.r.t. x , we get $\frac{dy}{dx} = \frac{1}{x}$.

$\therefore$ The area of the surface generated is

$$\begin{aligned}
S &= \int_A^B 2\pi x \, ds = 2\pi \int_{x_1}^{x_2} x \frac{ds}{dx} dx \\
&= 2\pi \int_1^e x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_1^e x \sqrt{1 + \frac{1}{x^2}} dx \\
&= 2\pi \int_1^e x \frac{\sqrt{x^2+1}}{x} dx = 2\pi \int_1^e \sqrt{1+x^2} dx \\
&= 2\pi \left[\frac{1}{2} x \sqrt{1+x^2} + \frac{1}{2} \log|x + \sqrt{1+x^2}| \right]_1^e \\
&= \pi [e\sqrt{1+e^2} + \log(e + \sqrt{1+e^2}) - 1\sqrt{1+1} - \log(1 + \sqrt{1+1})] \\
&= \pi [e\sqrt{1+e^2} + \log(e + \sqrt{1+e^2}) - \sqrt{2} - \log(1 + \sqrt{2})]. \quad \underline{\text{Ans.}}
\end{aligned}$$

3. The spherical surface with unit radius is generated by rotating the curve $x = \cos t$, $y = \sin t$ about x -axis, where t ranges over $[0, \pi]$. Find its area.

Solution. Given curve (unit semi-circle) is $x = \cos t$, $y = \sin t$, where $0 \leq t \leq \pi$.

Differentiating w.r.t. t , we get

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t$$

$$\therefore \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \sin^2 t + \cos^2 t = 1$$

$$\Rightarrow \frac{ds}{dt} = 1 \Rightarrow ds = dt.$$

$\therefore$ The area of the spherical surface generated is

$$\begin{aligned}
S &= \int_A^B 2\pi y \, ds = 2\pi \int_0^\pi \sin t \, dt \\
&= 2\pi [-\cos t]_0^\pi \\
&= 2\pi [-\cos \pi + \cos 0] = 2\pi [1 + 1] \\
&= 4\pi. \quad \underline{\text{Ans.}}
\end{aligned}$$

Remark: If the spherical surface has the radius r , it can be generated by rotating the curve (semi-circle) $x = r \cos t$, $y = r \sin t$, $0 \leq t \leq \pi$, about x -axis.

$$\text{So in this case, } \frac{dx}{dt} = -r \sin t, \quad \frac{dy}{dt} = r \cos t,$$

$$\left(\frac{ds}{dt}\right)^2 = r^2 \sin^2 t + r^2 \cos^2 t = r^2 = r^2 \Rightarrow \frac{ds}{dt} = r \Rightarrow ds = r dt.$$

$\therefore$ The area of the spherical surface of radius r is

$$S = \int_A^B 2\pi y ds = 2\pi \int_0^\pi r \sin t r dt = 2\pi r^2 [-\cos t]_0^\pi = 2\pi r^2 [1 + 1] \\ = 4\pi r^2. \quad \underline{\text{Ans.}}$$

4. Prove that the surface of the oblate spheroid formed by the revolution of the ellipse of the semi-major axis a and eccentricity e is $2\pi a^2 \left[1 + \frac{1-e^2}{2e} \log \left(\frac{1+e}{1-e}\right)\right]$.

Solution. The oblate spheroid is generated by the revolution of the ellipse about its minor axis. Equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b = a\sqrt{1-e^2}$, $e < 1$.

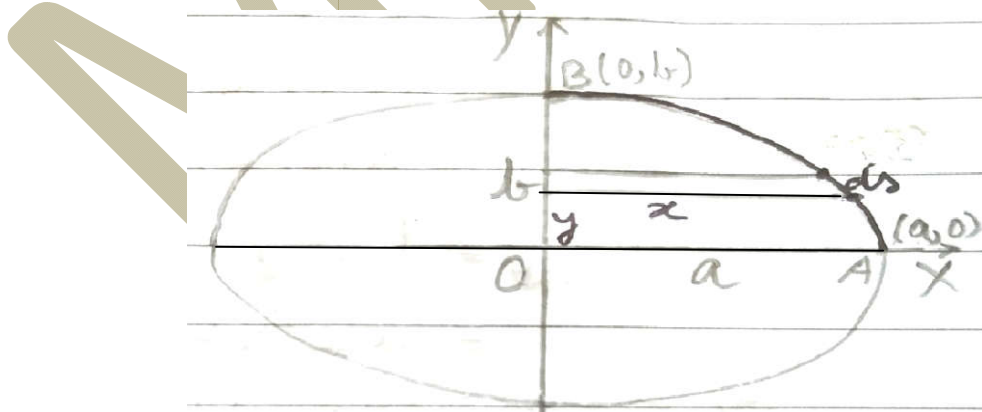
Let the parametric equations of the ellipse are

$x = a \cos t$, $y = b \sin t$, where t is parameter.

Differentiating w.r.t. t , $\frac{dx}{dt} = -a \sin t$, $\frac{dy}{dt} = b \cos t$

$$\therefore \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = a^2 \sin^2 t + b^2 \cos^2 t \\ = a^2 \sin^2 t + a^2 (1 - e^2) \cos^2 t = a^2 (1 - e^2 \cos^2 t)$$

$$\Rightarrow \frac{ds}{dt} = a\sqrt{1 - e^2 \cos^2 t} \Rightarrow ds = a\sqrt{1 - e^2 \cos^2 t} dt.$$



The ellipse is symmetrical about both the axes and for the arc of the ellipse lying in the first quadrant, t varies from 0 to $\pi/2$.

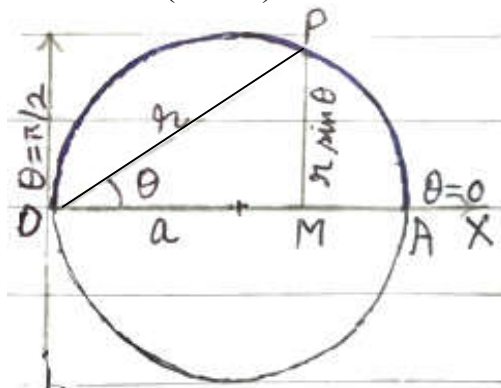
We have to revolve the ellipse about its minor axis, which is the y -axis.

$$\begin{aligned}
\therefore \text{The required area of the surface} &= 2 \int_A^B 2\pi x \, ds \\
&= 2 \times 2\pi \int_0^{\pi/2} a \cos t \cdot a \sqrt{1 - e^2 \cos^2 t} \, dt \\
&= 4\pi a^2 \int_0^{\pi/2} \sqrt{1 - e^2 + e^2 \sin^2 t} \cos t \, dt \\
&= 4\pi a^2 \int_0^e \sqrt{(1 - e^2) + z^2} \frac{dz}{e}, \text{ by } e \sin t = z \text{ so that } e \cos t \, dt = dz \\
&= \frac{4\pi a^2}{e} \int_0^e \sqrt{c^2 + z^2} \, dz, \text{ where } c = \sqrt{1 - e^2} \\
&= \frac{4\pi a^2}{e} \left[\frac{z}{2} \sqrt{c^2 + z^2} + \frac{c^2}{2} \log|z + \sqrt{c^2 + z^2}| \right]_0^e \\
&= \frac{4\pi a^2}{2e} \left[z\sqrt{1 - e^2 + z^2} + (1 - e^2) \log|z + \sqrt{1 - e^2 + z^2}| \right]_0^e \\
&= \frac{2\pi a^2}{e} \left[e\sqrt{1} + (1 - e^2) \log|e + \sqrt{1}| - 0 - (1 - e^2) \log|\sqrt{1 - e^2}| \right] \\
&= \frac{2\pi a^2}{e} \left[e + (1 - e^2) \log(e + 1) - (1 - e^2) \log \sqrt{1 - e^2} \right] \\
&= \frac{2\pi a^2}{e} \left[e + (1 - e^2) \log \left\{ \frac{(1+e)}{\sqrt{(1-e)(1+e)}} \right\} \right] = 2\pi a^2 \left[1 + \frac{(1-e^2)}{e} \log \sqrt{\frac{1+e}{1-e}} \right] \\
&= 2\pi a^2 \left[1 + \frac{1-e^2}{2e} \log \left(\frac{1+e}{1-e} \right) \right]. \quad \text{Proved.}
\end{aligned}$$

5. Find the area of the surface of revolution formed by revolving the curve $r = 2a \cos \theta$ about the initial line.

Solution. The given curve is $r = 2a \cos \theta$ (1)

It is a circle of radius a passing through pole (origin) and having diameter through the pole and on initial line (x-axis).



At pole O, $r = 0$ and $\theta = \pm \pi/2$, by (1).

At point A, $\theta = 0$ and $r = 2a$, by (1).

Differentiating (1) w.r.t. θ , we get

$$\frac{dr}{d\theta} = -2a \sin \theta \dots\dots\dots (2)$$

$$\therefore \left(\frac{ds}{d\theta}\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2 = 4a^2 \cos^2 \theta + 4a^2 \sin^2 \theta = 4a^2, \text{ by (1) \& (2)}$$

$$\Rightarrow \left(\frac{ds}{d\theta}\right)^2 = 4a^2 \Rightarrow \frac{ds}{d\theta} = 2a \dots\dots\dots (3)$$

The upper half curve (semi-circle APO) is revolved about initial line OX.

$$\begin{aligned} \therefore \text{The required area of the surface of revolution} &= \int_A^O 2\pi y \, ds \\ &= 2\pi \int_0^{\pi/2} r \sin \theta \frac{ds}{d\theta} d\theta = 2\pi \int_0^{\pi/2} 2a \cos \theta \sin \theta 2a d\theta, \text{ by (1) \& (3)} \\ &= 2\pi 2a^2 \int_0^{\pi/2} 2 \sin \theta \cos \theta d\theta = 4\pi a^2 \int_0^{\pi/2} \sin 2\theta d\theta \\ &= 4\pi a^2 \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2} = 4\pi a^2 \left[+\frac{1}{2} + \frac{1}{2} \right] \\ &= 4\pi a^2. \quad \text{Ans.} \end{aligned}$$

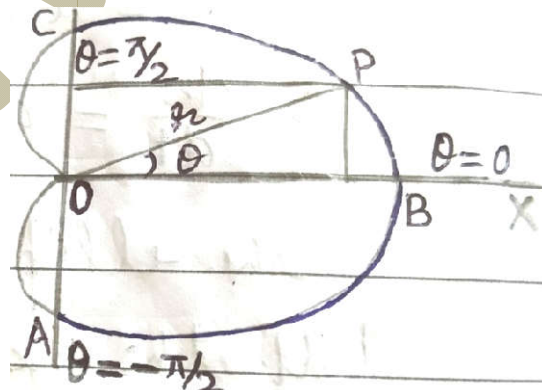
6. Find the surface area of the solid generated by the arc of the cardioid

$$r = a(1 + \cos \theta) \text{ included between } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

when rotated about (a) $\theta = 0$ and (b) $\theta = \pi/2$.

Solution. Given curve (cardioid) is

$$r = a(1 + \cos \theta).$$



Differentiating it w.r.t. θ , we get

$$\frac{dr}{d\theta} = a(0 - \sin \theta)$$

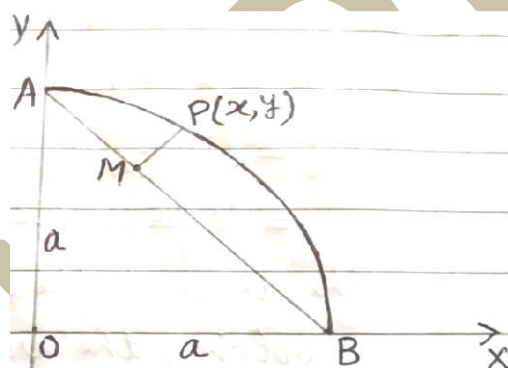
$$\therefore \left(\frac{ds}{d\theta}\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2 = a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta \text{ etc.}$$

Now try as above example 5.

7. The segment of the circle $x^2 + y^2 = a^2$ in the first quadrant cut off by the chord $x + y = a$ is revolved about the chord.

Find the area of the surface generated.

Solution. Try yourself.



LECTURE NO.:- 25

Volumes of Solids of Revolution

When a plane area revolves about a fixed line, it generates a solid, which is called the solid of revolution. The fixed line is called the axis of revolution or rotation.

For example, when a rectangular area revolves about one of its sides, it generates a solid cylinder; when a right angled triangle revolves about one of its perpendicular sides, it generates a right circular cone; when a semi circular area revolves about its bounding diameter, it generates a solid sphere.

Here following cases may be:

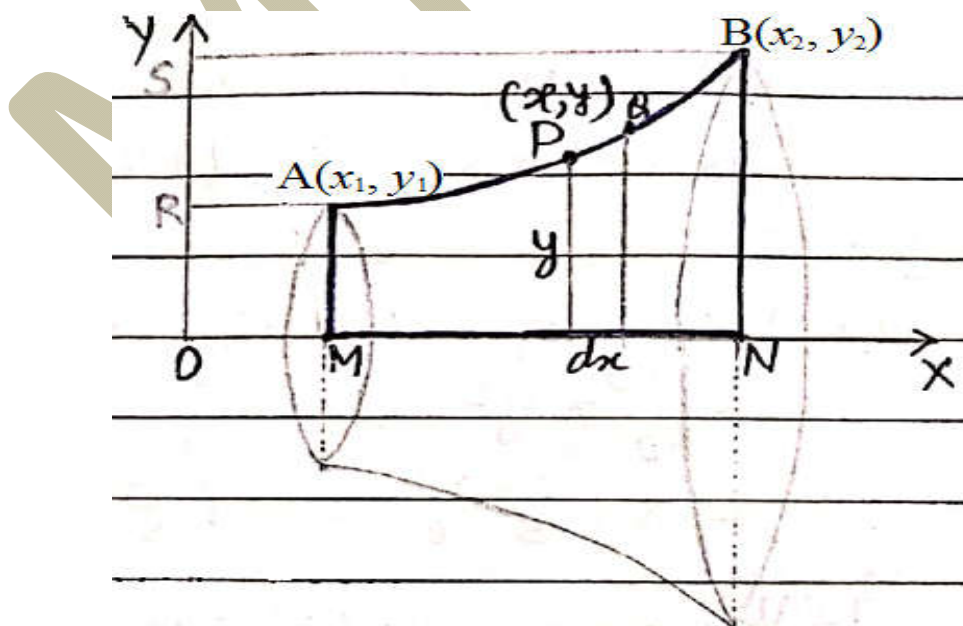
1. When the curve is given in cartesian form $y = f(x)$ or $x = F(y)$:

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two points on the curve $y = f(x)$ or $x = F(y)$.

(i) Revolution about x -axis:

The volume of the solid generated by the revolution of the area bounded by the curve $y = f(x)$, x -axis and lines $x = x_1$ & $x = x_2$ (the lines passing through A & B perpendicular to x -axis), about x -axis is given by

$$V = \int_A^B \pi y^2 dx = \pi \int_{x_1}^{x_2} [f(x)]^2 dx, \because \text{elementary volume} = \pi y^2 dx.$$



Alternatively, if the equation of the curve is $x = F(y)$, then

$$V = \int_A^B \pi y^2 dx = \pi \int_{y_1}^{y_2} y^2 \frac{dx}{dy} dy = \pi \int_{y_1}^{y_2} y^2 F'(y) dy.$$

(ii) Revolution about y-axis:

The volume of the solid generated by the revolution of the area bounded by the curve $y = f(x)$, y -axis and lines $y = y_1$ & $y = y_2$ (the lines passing through A & B perpendicular to y -axis), about y -axis is given by

$$V = \int_A^B \pi x^2 dy = \pi \int_{x_1}^{x_2} x^2 \frac{dy}{dx} dx = \pi \int_{x_1}^{x_2} x^2 f'(x) dx.$$

Alternatively, if the equation of the curve is $x = F(y)$, then

$$V = \int_A^B \pi x^2 dy = \pi \int_{y_1}^{y_2} [F(y)]^2 dy.$$

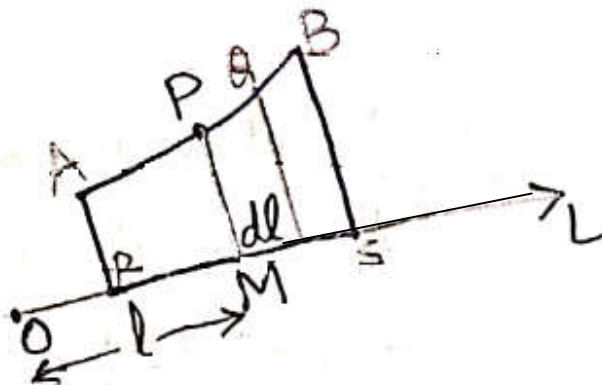
(iii) Revolution about any line, which is one side of the area of revolution:

The volume of the solid generated by the revolution of the area bounded by the curve $y = f(x)$, given line L and lines passing through A & B perpendicular to the line L, about line L is given by

$$V = \int_A^B \pi (PM)^2 dl = \pi \int_{x_1}^{x_2} (PM)^2 \frac{dl}{dx} dx,$$

where P is any point (x, y) on the arc AB of the curve, PM is the length of the perpendicular on line L from the point P and dl is elementary length on line L so that

$$\frac{dl}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \text{ on line L.}$$



$$\text{Alternatively, if } x = F(y), \text{ then } V = \pi \int_{y_1}^{y_2} (PM)^2 \frac{dl}{dy} dy, \frac{dl}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \text{ on L.}$$

(iv) Revolution about any line, which is not a side of the area of revolution:

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve $y = f(x)$ and other curve $y = g(x)$ or given line(s), about given line L is given by

$$V = \int_A^B \pi(h^2 - k^2) dl = \pi \int_{x_1}^{x_2} (h^2 - k^2) \frac{dl}{dx} dx,$$

where h and $k (< h)$ are heights of perpendiculars to the line L from any point (x, y) on the arc AB and from corresponding point on opposite boundary respectively and dl is elementary length on line L so that $\frac{dl}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ on line L.

$$\text{Alternatively, if } x = F(y), V = \pi \int_{y_1}^{y_2} (h^2 - k^2) \frac{dl}{dy} dy, \frac{dl}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \text{ on L.}$$

(v) Revolution about line $y = k$ (parallel to x-axis): Volume of revolution is

$$V = \int_A^B \pi(y - k)^2 dx = \pi \int_{x_1}^{x_2} [f(x) - k]^2 dx, \text{ if } y = f(x).$$

$$\text{Alternatively, } V = \pi \int_{y_1}^{y_2} (y - k)^2 \frac{dx}{dy} dy = \pi \int_{y_1}^{y_2} (y - k)^2 F'(y) dy, \text{ if } x = F(y).$$

(vi) Revolution about line $x = h$ (parallel to y-axis): If $y = f(x)$, then volume is

$$V = \int_A^B \pi(x - h)^2 dy = \pi \int_{x_1}^{x_2} (x - h)^2 \frac{dy}{dx} dx = \pi \int_{x_1}^{x_2} (x - h)^2 f'(x) dx.$$

$$\text{Alternatively, if } x = F(y), \text{ then } V = \pi \int_A^B (x - h)^2 dy = \pi \int_{y_1}^{y_2} [F(y) - h]^2 dy.$$

2. When the curve is given in parametric form $x = x(t), y = y(t)$:

Let $A(t_1)$ and $B(t_2)$ be two points on the curve $x = x(t), y = y(t)$.

(i) Revolution about x-axis:

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve, x-axis and lines through A & B perpendicular to x-axis, about x-axis is given by

$$V = \int_A^B \pi y^2 dx = \pi \int_{t_1}^{t_2} [y(t)]^2 \frac{dx}{dt} dt.$$

(ii) Revolution about y-axis:

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve, y-axis and lines through A & B perpendicular to y-axis, about y-axis is given by

$$V = \int_A^B \pi x^2 dy = \pi \int_{t_1}^{t_2} [x(t)]^2 \frac{dy}{dt} dt.$$

(iii) Revolution about any line, which is one side of the area of revolution:

The volume of the solid generated by the revolution of the area bounded by the arc AB of the curve, given line L and lines passing through A & B perpendicular to the line L, about line L is given by

$$V = \int_A^B \pi (PM)^2 dl = \pi \int_{t_1}^{t_2} (PM)^2 \frac{dl}{dt} dt,$$

where P is any point on the arc AB of the curve, PM is the length of the perpendicular on line L from the point P and dl is elementary length on line L so that

$$\frac{dl}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \text{ on line L.}$$

(iv) Revolution about any line, which is not a side of the area of revolution:

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve and other curve or given line(s), about given line L is given by

$$V = \int_A^B \pi (h^2 - k^2) dl = \pi \int_{t_1}^{t_2} (h^2 - k^2) \frac{dl}{dt} dt,$$

where h and $k (< h)$ are heights of perpendiculars to the line L from any point (t) on the arc AB and from corresponding point on opposite boundary respectively and dl is the elementary length on line L so that $\frac{dl}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ on line L.

(v) Revolution about lines parallel to coordinate axis:

Putting $x = x(t)$, $y = y(t)$ and corresponding values in the formulae of (v) and (vi), the volumes of revolutions can be obtained.

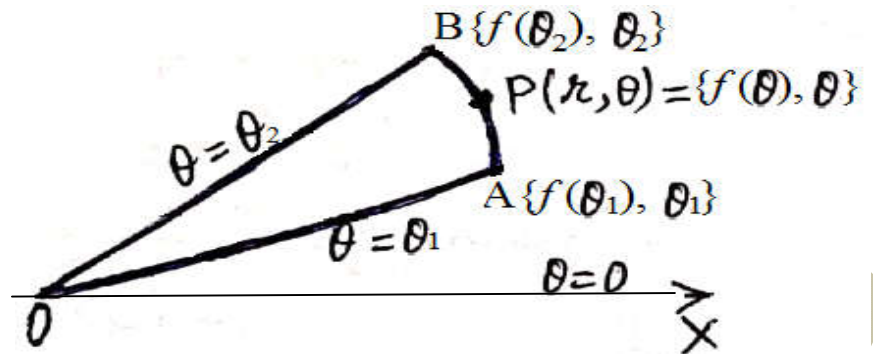
3. When the curve is given in polar form $r = f(\theta)$ or $F(r, \theta) = 0$:

Let $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ be two points on the curve $r = f(\theta)$ or $F(r, \theta) = 0$.

(i) Revolution about initial line $\theta = 0$ (x-axis):

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve and lines $\theta = \theta_1$ & $\theta = \theta_2$, about initial line $\theta = 0$ (x-axis) is given by

$$V = \int_A^B \pi(r \sin \theta)^2 d(r \cos \theta) = \int_{\theta_1}^{\theta_2} \frac{2}{3} \pi r^3 \sin \theta d\theta = \frac{2}{3} \pi \int_{\theta_1}^{\theta_2} [f(\theta)]^3 \sin \theta d\theta.$$

**(ii) Revolution about line through the pole perpendicular to the initial line i.e. about line $\theta = \pi/2$ (y-axis):**

The volume of the solid generated by the revolution of the area bounded by arc AB of the curve and lines $\theta = \theta_1$ & $\theta = \theta_2$, about line $\theta = \pi/2$ (y-axis) is given by

$$V = \int_A^B \pi(r \cos \theta)^2 d(r \sin \theta) = \int_{\theta_1}^{\theta_2} \frac{2}{3} \pi r^3 \cos \theta d\theta = \frac{2}{3} \pi \int_{\theta_1}^{\theta_2} [f(\theta)]^3 \cos \theta d\theta.$$

(iii) Revolution about any line $\theta = \alpha$:

Replacing θ by $\theta + \alpha$ in the whole system (equations of the curve and lines), the given line $\theta = \alpha$ becomes initial line $\theta = 0$. Then we can use (i).

LECTURE NO.:- 26

Problems on Volumes of Solids of Revolution

1. A (3, 0) be a point on x-axis, B (0, 9) be a point on y-axis and C (3, 9) be a point on parabola $y = x^2$. Find the volume of solid of revolution generated when the area OAC, bounded by $y = x^2$, $x = 3$ and x- axis is rotated about
(a) x- axis, (b) y- axis, (c) line AC, (d) line BC.

Solution. Given curve is parabola $y = x^2 \dots (1)$

- (a) Volume of solid of revolution of area OAC

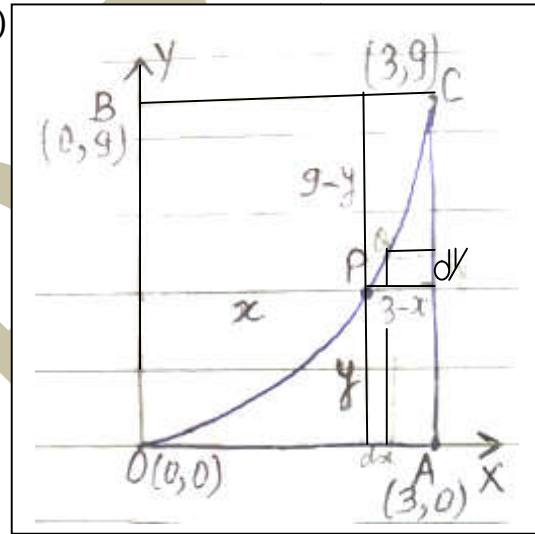
$$\text{about x- axis} = \int_0^A \pi y^2 dx$$

$$= \pi \int_0^3 [x^2]^2 dx, \text{ by (1)}$$

$$= \pi \left[\frac{x^5}{5} \right]_0^3$$

$$= \pi \left[\frac{3^5}{5} - 0 \right]$$

$$= \frac{243\pi}{5}. \text{ Ans.}$$



- (b) Volume of solid of revolution of area OAC about y- axis

$$= \int_A^C \pi(3^2 - x^2) dy = \pi \int_0^9 (9 - y) dy, \text{ by (1)}$$

$$= \pi \left[9y - \frac{y^2}{2} \right]_0^9 = \pi \left[9 \times 9 - \frac{9^2}{2} \right]$$

$$= \pi \left(81 - \frac{81}{2} \right) = \pi \left(\frac{81}{2} \right)$$

$$= \frac{81\pi}{2}. \text{ Ans.}$$

- (c) Volume of solid of revolution of area OAC about line AC = $\int_A^C \pi(3 - x)^2 dy$

$$= \pi \int_0^9 (3 - \sqrt{y})^2 dy, \text{ by (1)}$$

$$= \pi \int_0^9 (9 - 6\sqrt{y} + y) dy = \pi \left[9y - 6 \frac{y^{3/2}}{3/2} + \frac{y^2}{2} \right]_0^9$$

$$= \pi \left[9 \times 9 - 2 \times 2(9)^{3/2} + \frac{9^2}{2} - \{0\} \right] = \pi \left[81 - 4 \times 27 + \frac{81}{2} \right]$$

$$= 27\pi \left[3 - 4 + \frac{3}{2} \right] = 27\pi \left[\frac{1}{2} \right]$$

$$= \frac{27\pi}{2} . \quad \underline{\text{Ans.}}$$

(d) Volume of solid of revolution of area OAC about line BC

$$= \int_0^4 \pi [9^2 - (9 - y)^2] dx = \int_0^3 \pi [9^2 - (9 - x^2)^2] dx, \text{ by (1)}$$

$$= \pi \int_0^3 [9^2 - (9^2 - 18x^2 + x^4)] dx = \pi \int_0^3 [18x^2 - x^4] dx$$

$$= \pi \left[18 \frac{x^3}{3} - \frac{x^5}{5} \right]_0^3 = \pi \left[6 \times 3^3 - \frac{3^5}{5} - \{0\} \right]$$

$$= 3^4 \pi \left(2 - \frac{3}{5} \right) = 81\pi \left(\frac{7}{5} \right)$$

$$= \frac{567\pi}{5} . \quad \underline{\text{Ans.}}$$

2. Determine the volume of solid generated by revolving the plane area bounded by parabola $y^2 = 4x$ and $x = 4$ about the line $x = 4$.

Solution. Given curve is parabola

$$y^2 = 4x \dots\dots\dots (1)$$

We have to determine the volume of solid generated by revolving the plane area ABOA about line AB.

$\therefore$ Required volume of solid

$$= \int_B^A \pi (4 - x)^2 dy$$

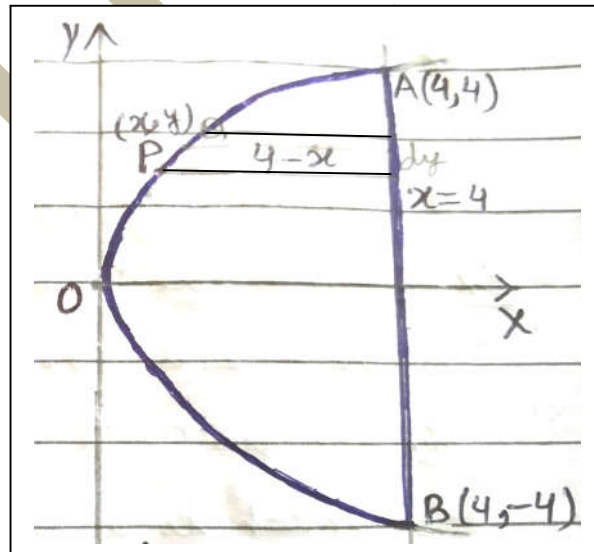
$$= \pi \int_{-4}^4 \left(4 - \frac{y^2}{4} \right)^2 dy, \text{ by (1)}$$

$$= \pi \int_{-4}^4 \left(16 - 2y^2 + \frac{y^4}{16} \right) dy$$

$$= \pi 2 \int_0^4 \left(16 - 2y^2 + \frac{y^4}{16} \right) dy, \text{ by a property of definite integral}$$

$$= 2\pi \left[16y - 2 \frac{y^3}{3} + \frac{y^5}{16 \times 5} \right]_0^4 = 2\pi \left[16 \times 4 - 2 \times \frac{4^3}{3} + \frac{4^5}{16 \times 5} - \{0\} \right]$$

$$= 2\pi 4^3 \left[1 - \frac{2}{3} + \frac{1}{5} \right] = 128\pi \left[\frac{15 - 10 + 3}{15} \right] = 128\pi \left[\frac{8}{15} \right]$$



$$= \frac{1024\pi}{15} \cdot \underline{\text{Ans.}}$$

3. Calculate the volume of the solid of revolution generated by revolving the hypocycloid $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$ (a) about x -axis, (b) about y -axis.

Deduce the results when asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ is revolved (c) about x -axis, (d) about y -axis.

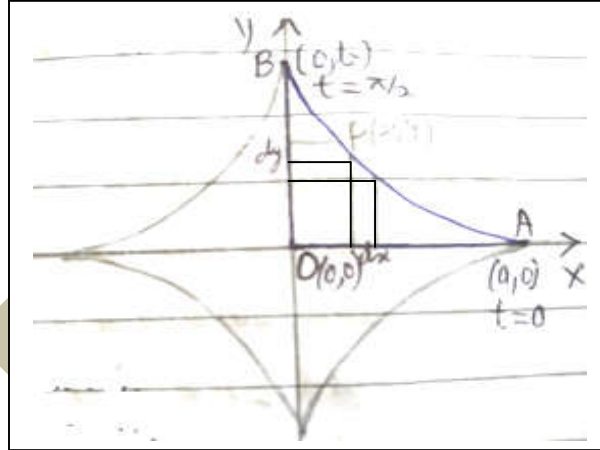
Solution. Given curve (hypocycloid) is

$$\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1 \dots\dots\dots (1)$$

Let us take parametric equations

$$x = a \cos^3 t, y = b \sin^3 t \dots (2)$$

Here t is parameter.



Curve is symmetrical about both the axes.

- (a) Volume of the solid of revolution by revolving curve (1) about x -axis

$$= 2 \int_0^A \pi y^2 dx = 2 \int_0^a \pi y^2 dx$$

$$= 2 \int_{t_1}^{t_2} \pi y^2 \frac{dx}{dt} dt$$

$$= 2\pi \int_{\pi/2}^0 [b \sin^3 t]^2 \frac{d}{dt} (a \cos^3 t) dt, \text{ by (2), } \therefore \text{ limit}$$

x	0	a
t	$\pi/2$	0

$$= 2\pi \int_{\pi/2}^0 b^2 \sin^6 t \cdot a 3 \cos^2 t (-\sin t) dt$$

$$= 2\pi b^2 a 3 \int_0^{\pi/2} \sin^7 t \cdot \cos^2 t dt, \text{ using a property of definite integral}$$

$$= 6\pi ab^2 \frac{\Gamma\left(\frac{7+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2 \Gamma\left(\frac{7+2+2}{2}\right)}, \text{ by gamma function formula}$$

$$= 6\pi ab^2 \frac{\Gamma_4 \Gamma\left(\frac{3}{2}\right)}{2 \Gamma\left(\frac{11}{2}\right)} = 6\pi ab^2 \frac{3! \times \frac{1}{2} \sqrt{\pi}}{2 \times \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}, \text{ by } \Gamma(n+1) = n \Gamma n$$

$$= \frac{32\pi ab^2}{105} \cdot \underline{\text{Ans.}}$$

(b) By symmetry, the volume of the solid of revolution by revolving curve (1) about y -axis

$$= \frac{32\pi a^2 b}{105} \cdot \underline{\text{Ans.}}$$

(c) In case of asteroid, putting $b = a$ in (a), the required volume of the solid of revolution about x -axis $= \frac{32\pi a^3}{105} \cdot \underline{\text{Ans.}}$

(d) Similarly, putting $b = a$ in (c), the required volume of the solid of revolution about y -axis $= \frac{32\pi a^3}{105} \cdot \underline{\text{Ans.}}$

4. Find the volume of solid generated by the revolution of $r = 2a \cos \theta$ about the initial line.

Solution. Given curve is

$$r = 2a \cos \theta \dots\dots\dots (1)$$

It is a circle of radius a passing through pole (origin) and having diameter through the pole as initial line (x -axis).

On initial line, $\theta = 0$, $r = 2a$, by (1).

On pole O, $r = 0$, $\theta = \pi/2$, by (1).

If we replace θ by $-\theta$ in (1), then

the equation of the curve remains unchanged.

So the curve is symmetrical about initial line (x -axis).

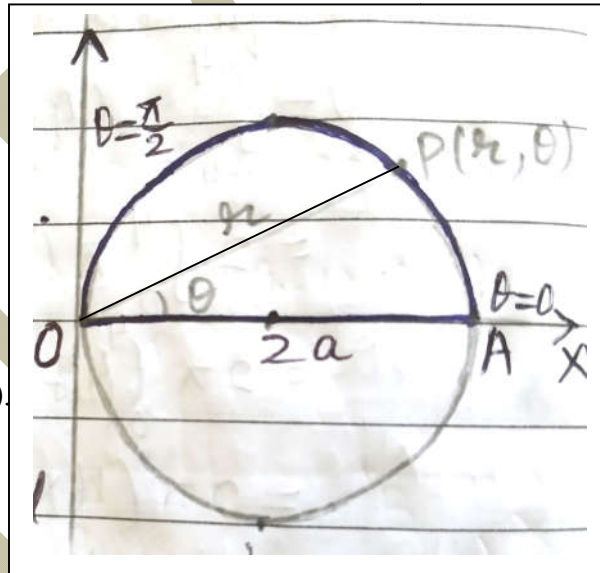
In the first quadrant, θ varies from 0 (at A) to $\pi/2$ (at O).

$\therefore$ The volume of the solid of revolution of the curve (1) about initial line is

$$V = \int_A^O \frac{2}{3} \pi r^3 \sin \theta \, d\theta = \frac{2}{3} \pi \int_0^{\pi/2} [2a \cos \theta]^3 \sin \theta \, d\theta, \text{ by (1)}$$

$$= \frac{2}{3} \pi 8a^3 \int_0^{\pi/2} \cos^3 \theta \cdot \sin \theta \, d\theta = \frac{16}{3} \pi a^3 \left[-\frac{\cos^4 \theta}{4} \right]_0^{\pi/2} = \frac{16}{3} \pi a^3 \left[0 + \frac{1}{4} \right]$$

$$= \frac{4}{3} \pi a^3 \cdot \underline{\text{Ans.}}$$



5. Find the volume generated when the plane area bounded by the parabola $y + x^2 + 3x - 6 = 0$ and the line $x + y = 3$ is revolved about the line.

Solution. Given parabola is $y + x^2 + 3x - 6 = 0$ (1)

$$\Rightarrow \left(x + \frac{3}{2}\right)^2 = -\left(y - \frac{33}{4}\right); \text{Vertex } \left(-\frac{3}{2}, \frac{33}{4}\right), \text{latus rectum} = 1.$$

Given line is $x + y = 3$ (2). Diff, $dx + dy = 0$

The points of intersection of (1), (2) & axes are shown in fig.

$$PM = \frac{x+y-3}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}}(x+y-3).$$

$$MN = dl = \sqrt{dx^2 + dy^2} = \sqrt{dx^2 + (-dx)^2} = \sqrt{2} dx.$$

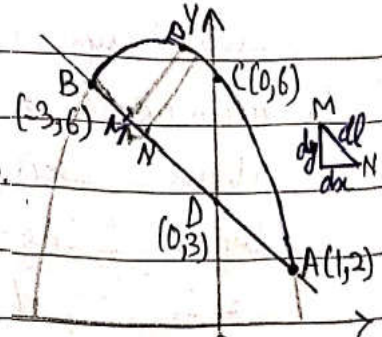
The required volume of revolution (say V) is generated by plane area ACBDA about the line BDA. So $V = \int_B^A \pi PM^2 dl = \int_{-3}^1 \pi \left(\frac{x+y-3}{\sqrt{2}}\right)^2 \sqrt{2} dx$

$$\Rightarrow V = \frac{\pi\sqrt{2}}{2} \int_{-3}^1 (x+y-3)^2 dx = \frac{\pi}{\sqrt{2}} \int_{-3}^1 [x + (-x^2 - 3x + 6) - 3]^2 dx, \text{ by (1)}$$

$$= \frac{\pi}{\sqrt{2}} \int_{-3}^1 [-x^2 - 2x + 3]^2 dx = \frac{\pi}{\sqrt{2}} \int_{-3}^1 [x^4 + 4x^3 - 6x^2 + 4x^2 - 12x + 9] dx$$

$$= \frac{\pi}{\sqrt{2}} \left[\frac{1}{5} x^5 + x^4 - \frac{2}{3} x^3 - 6x^2 + 4x^2 - 12x + 9 \right]_{-3}^1 = \frac{\pi}{\sqrt{2}} \left[\frac{1}{5} + 1 - \frac{2}{3} - 6 + 9 + \frac{243}{5} - 81 - 18 + 54 + 27 \right]$$

$$= \frac{\pi}{\sqrt{2}} \left[\frac{244}{5} - \frac{2}{3} - 14 \right] = \frac{\pi}{\sqrt{2} \cdot 15} (732 - 10 - 210) = \frac{\pi \cdot 256 \times 2}{\sqrt{2} \cdot 15} = \frac{256\sqrt{2} \pi}{15} \text{ Ans.}$$



6. The cardioid $r = a(1 + \cos \theta)$ revolves about the initial line. Find the volume of the solid thus generated.

Solution. Try yourself.